Case Study Report

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Cloud Services and Deployment Models: A Cross-Sector Case Study Analysis

1. Introduction

Cloud computing has fundamentally reshaped how organizations across every sector deliver services, manage infrastructure, and drive automation. Rather than owning and operating physical data centers, enterprises now consume compute, storage, networking, and intelligence on demand - scaling elastically, paying only for what they use, and offloading infrastructure management to specialist providers (Mell & Grance, 2011). The shift from Capital Expenditure (CAPEX)-heavy on-premises environments to Operational Expenditure (OPEX)-driven cloud models is no longer a trend exclusive to technology companies - it is a cross-industry imperative that spans government agencies, AI-native startups, and century-old industrial manufacturers alike (Linthicum, 2021; McHaney, 2021).

This report analyses three real-world case studies drawn from distinct sectors – government law enforcement (Victoria Police), artificial intelligence and software startup (ChatAndBuild), and industrial manufacturing (Voith) – each provided by the subject facilitator and representing a different combination of cloud service model, deployment model, and provider ecosystem. By examining how cloud was applied in each context, this report aims to: describe the essential cloud elements that enabled business automation in each case; distinguish cloud adoption from traditional IT infrastructure approaches and identify the key cloud service offerings that made each outcome possible.

The three cases collectively illustrate how cloud computing is not a single, monolithic technology but a spectrum of services and deployment strategies calibrated to organizational

context, risk appetite, and automation goals. Victoria Police leveraged Microsoft Power Platform (PaaS) within a hybrid multi-environment model to democratize application development across a law enforcement agency – transforming previously invisible, ungoverned Microsoft Access databases into a governed, scalable innovation ecosystem (Microsoft, 2026). ChatAndBuild, an AI-native startup based in Singapore, migrated to Google Cloud Platform to eliminate the manual infrastructure burden that was impeding real-time AI agent creation, achieving a 45% reduction in infrastructure overhead and a 30-70% improvement in generation latency (Google Cloud, n.d.). Voith, a 150-year-old manufacturing company operating across 60 countries, selected AWS following a competitive provider assessment to consolidate 3,500 servers distributed across 146 global locations into six AWS regions – projecting 30% cost savings compared to the legacy setup (Amazon Web Services, n.d.).

The following sections examine each case in turn - analyzing service and deployment model choices, mapping specific cloud services to their automation outcomes, and reflecting on the contrast with traditional IT approaches - before synthesizing the findings in a concluding analysis.

2. Case Study 1 – Victoria Police & Microsoft Power Platform

2.1. Case Overview

Victoria Police – Australia's principal state law enforcement agency – adopted Microsoft Power Platform to establish a **fusion development framework**: a governed model enabling non-professional developers (police personnel across business units) to build data-driven applications safely, without depending on a constrained central IT team (Microsoft, 2026).

The catalyst was a pandemic-era surge in self-directed app-building, when employees teaching themselves Power Apps revealed both the appetite for low-code tooling and the risk of ungoverned 'shadow' applications built on standalone Microsoft Access databases. Rather than restricting this behavior, Victoria Police channeled it – creating structured pathways from personal productivity through to enterprise-grade production, with two fusion-born applications already progressing to enterprise adoption (Microsoft, 2026).

2.2. Service Model Analysis

Victoria Police's cloud adoption aligns primarily with a **Platform as a Service (PaaS)** model. Power Apps and Power Platform abstract the underlying infrastructure – Microsoft manages the compute, networking, and database layers of Dataverse and the Power Platform runtime – while Victoria Police's developers focus on application logic, data models, and user experience (McHaney, 2021). This is the defining characteristic of PaaS: the provider controls the platform, the customer controls the application (IBM, n.d.0a). Some Microsoft 365 productivity workloads (SharePoint, Teams, Outlook) sit at the **SaaS** layer – fully managed software delivered over the network. There is no IaaS footprint in this case; Victoria Police has not provisioned raw virtual machines or unmanaged storage – all development occurs within the abstracted Power Platform environment (Microsoft, 2026).

Service Model	Characteristic	Fit for Victoria Police
IaaS	Customer manages OS, runtime, storage	Not applicable – no raw compute provisioned
PaaS	Provider manages platform; customer builds applications	**Primary model** – Power Apps, Dataverse, Power Platform runtime

SaaS	Provider manages everything; customer uses software	Partial – Microsoft 365 (SharePoint, Teams, Outlook) as data and collaboration substrate

Table 1: Service model analysis for Victoria Police (Microsoft, 2026).

2.3. Deployment Model Comparison

Victoria Police operates a **hybrid, multi-environment deployment model**: a personal productivity environment (the default Microsoft 365 tenant available to all staff), dedicated fusion environments (isolated, governed Power Platform environments provisioned on request by business units with executive endorsement), and enterprise production environments (reviewed and approved by the central Power Platform team) (Microsoft, 2026). This tiered model is a practical application of hybrid cloud architecture – multiple environment tiers serving different governance, security, and scale requirements simultaneously (Manvi & Shyam, 2021).

Deployment Model	Governance	Security	Scalability	Fit for Victoria Police
On-Premises (Microsoft Access)	Low – invisible to IT	High risk – unaudited shadow apps	None	Prior state – ungoverned shadow apps
Hybrid (Multi-env Power Platform)	High – tiered approval flow with exec endorsement	High – centralized standards, security review	Moderate	**Adopted** – governed innovation pathway
Public Cloud (ungoverned)	Low	Low	High	Same shadow IT risk as before
Private Cloud	High	High	Limited	High cost, no low-code ecosystem

Table 2: Deployment model comparison for Victoria Police (Microsoft, 2026; Manvi & Shyam, 2021).

2.4. Cloud Services Analysis

The fusion framework relies on a layered set of Microsoft Power Platform services, each enabling a distinct stage of the application lifecycle (Microsoft, 2026):

Service	Purpose	Business Outcome
Power Apps (Low-code / Paas)	Enable non-professional developers to build data-driven apps without code	Personnel build and iterate solutions without the central IT bottleneck
Microsoft Dataverse	Model-driven, structure data store underpinning enterprise apps	Replaces isolated MS Access databases with a governed, transparent and auditable data layer
Power Platform Environments	Isolated development, test, and production workspaces per business unit	Clear pathway from personal innovation to enterprise-grade development
Power Platform Approval Flow	Governs requests for dedicated fusion environments via a model-driven app	Ensures executive endorsement and security team review before promotion to production
Microsoft 365 (SharePoint)	Data substrate and collaboration layer for initial prototypes	Familiar toolset lowers the barrier to entry for non-technical builders
Copilot Studio (exploration)	AI-assisted app development using generative coding	Future capability to further accelerate the build-to-deploy cycle

Table 3: Microsoft Power Platform services utilized by Victoria Police and their business outcomes (Microsoft, 2026).

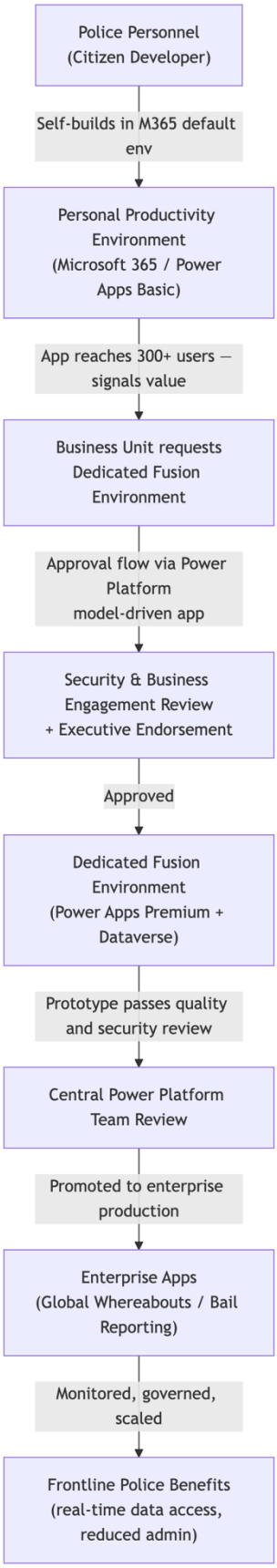
Police Personnel (Citizen Developer)
Self-builds in M365 default env
Personal Productivity Environment (Microsoft 365 / Power Apps Basic)
App reaches 300+ users — signals value
Business Unit requests Dedicated Fusion Environment
Approval flow via Power Platform model-driven app
Security & Business Engagement Review + Executive Endorsement
Approved
Dedicated Fusion Environment (Power Apps Premium + Dataverse)
Prototype passes quality and security review
Central Power Platform Team Review
Promoted to enterprise production
Enterprise Apps (Global Whereabouts / Bail Reporting)
Monitored, governed, scaled
Frontline Police Benefits (real-time data access, reduced admin)

Figure 1: Victoria Police fusion development lifecycle: from citizen developer prototype through tiered governance to enterprise production (Microsoft, 2026).

2.5. Reflection

The Victoria Police case demonstrates cloud's **on-demand self-service** characteristic (Mell & Grance, 2011) applied beyond technical teams – extending it to non-professional developers under tiered governance. In a traditional IT model, every application request queued behind a constrained central team, driving staff toward ungoverned Microsoft Access solutions outside IT visibility. Power Platform inverts this: it governs the _pathway_ to production while allowing distributed development (Microsoft, 2026) — a structural distinction that on-premises tooling cannot replicate (McHaney, 2021). The primary risk is data residency: sensitive law enforcement data in a PaaS environment requires compliance with the Australian Protective Security Policy Framework (PSPF), and the risk surface expands as adoption scales to thousands of officers across regions — a challenge Efrain Tionko (Head of Digital Delivery, Victoria Police) explicitly acknowledged (Microsoft, 2026).

3. Case Study 2 – ChatAndBuild & Google Cloud

3.1. Case Overview

ChatAndBuild is a Singapore-based AI-native startup whose platform enables anyone to turn natural language into working applications and AI agents instantly, serving over 140,000 users globally (Google Cloud, n.d.). The company's previous infrastructure imposed 30–70%

higher latency on real-time generation and caused context rot in long-lived AI agents – loss of user history that eroded platform trust. Global hackathon events required engineers to manually provision servers days in advance to prevent crashes from IP surges, making elasticity a critical business requirement. ChatAndBuild migrated to Google Cloud to establish **a fully AI-native, elastically scalable infrastructure** capable of operating at global speed (Google Cloud, n.d.).

3.2. Service Model Analysis

ChatAndBuild's architecture spans **IaaS** and **PaaS** layers. Google Compute Engine provides the raw virtual machine infrastructure (IaaS) that enables elastic scaling of the worker fleet from approximately 10 to 100+ instances during global hackathon events, handled automatically through instance groups and autoscaling policies (Google Cloud, n.d.). On top of this, a rich suite of managed PaaS services handles AI orchestration, containerized workload management, data analytics, state synchronization, and CI/CD (Continuous Integration / Continuous Development) – abstracting operational complexity so that the engineering team can focus on product development rather than infrastructure management (McHaney, 2021).

Service Model	Characteristic	Fit for Victoria Police
IaaS	Raw compute provisioned on demand; customer manages OS and runtime	Google Compute Engine – elastic worker fleetr scaling for burst demand
PaaS	Managed platform; provider handles runtime and infrastructure operations	**Primary layer** – Vertex AI, GKE, Cloud Run, BigQuery, Cloud SQL, Firebase
SaaS	Fully managed software	Partial – Firebase and Firestore as managed backend services

Table 4: Service model analysis for ChatAndBuild (Google Cloud, n.d.).

3.3. Deployment Model Comparison

ChatAndBuild operates on **public cloud** across multiple Google Cloud regions – a model in which cloud infrastructure is owned and managed by a third-party provider and delivered over the internet on a consumption basis (IBM, n.d.-b). Cloud Load Balancing routes each request to the nearest available region, ensuring sub-second response times globally. The multi-region model was the only viable option for a startup serving over 140,000 users across time zones, where hackathon spikes require instantly elastic capacity and prior on-premises approaches demanded days of manual server provisioning (Google Cloud, n.d.).

Deployment Model	Elasticity	Global Reach	Latency	Fit for ChatAndBuild
On-Premises	None – fixed servers	Single geography	High	Prior state – manual-provisioning; 30-70% higher latency; engineers as *"human autoscalers"*
Private Cloud	Limited	Single geography	Medium	No global edge; cannot serve 140,000+ users across time zones
Public Cloud (Multi-region)	High – Auto-scale 10→100+ instances in minutes	Global PoP via Cloud Load Balancing	Low – sub-second	**Adopted** – GKE + Cloud Run auto-scale to demand with zero downtime
Multi-Cloud	Highest	Highest	Lowest	Premature complexity for current startup scale

Table 5: Deployment model comparison for ChatAndBuild (Google Cloud, n.d.; Manvi & Shyam, 2021).

3.4. Cloud Services Analysis

The Google Cloud architecture forms an integrated "idea-to-app" engine where each service layer contributes to the real-time generation pipeline (Google Cloud, n.d.):

Service	Purpose	Business Outcome
Vertex AI + Gemini models	Routes prompts to specialized coding and research agents; multimodal generation	Core AI engine – 30–70% latency reduction vs previous setup
Google Kubernetes Engine (GKE)	Isolated namespaces per agent graph; manages thousands of concurrent builder sessions	Zero-downtime event scaling – 13B+ tokens processed at a global hackathon
Cloud Run + Cloud Functions	Stateless microservices that auto-spin for traffic bursts	Handles demand spikes without idle compute; no pre-provisioning required
Cloud Load Balancing	Routes every request to the nearest region	Sub-second response times globally
BigQuery	Telemetry layer – tracks billions of tokens to prevent context rot	Maintains long-term agent memory continuity and intelligence
Cloud SQL + Firestore + Firebase	Structured data management and real-time user state synchronization	Eliminates context rot; persistent, coherent agent sessions
Cloud CDN + Cloud Storage	Stores and delivers generated application assets globally	Final app delivery to end users with low latency
Cloud Build + Artifact Registry	CI/CD pipeline for continuous feature deployment	Feature delivery velocity doubled (2× faster)
VPC Service Controls + Cloud Armor + IAM	Security perimeter for regulated enterprise and government markets	SOC2 compliance; strict audit logging for enterprise and government partners

Table 6: Google Cloud services utilized by ChatAndBuild and their business outcomes (Google Cloud, n.d.).

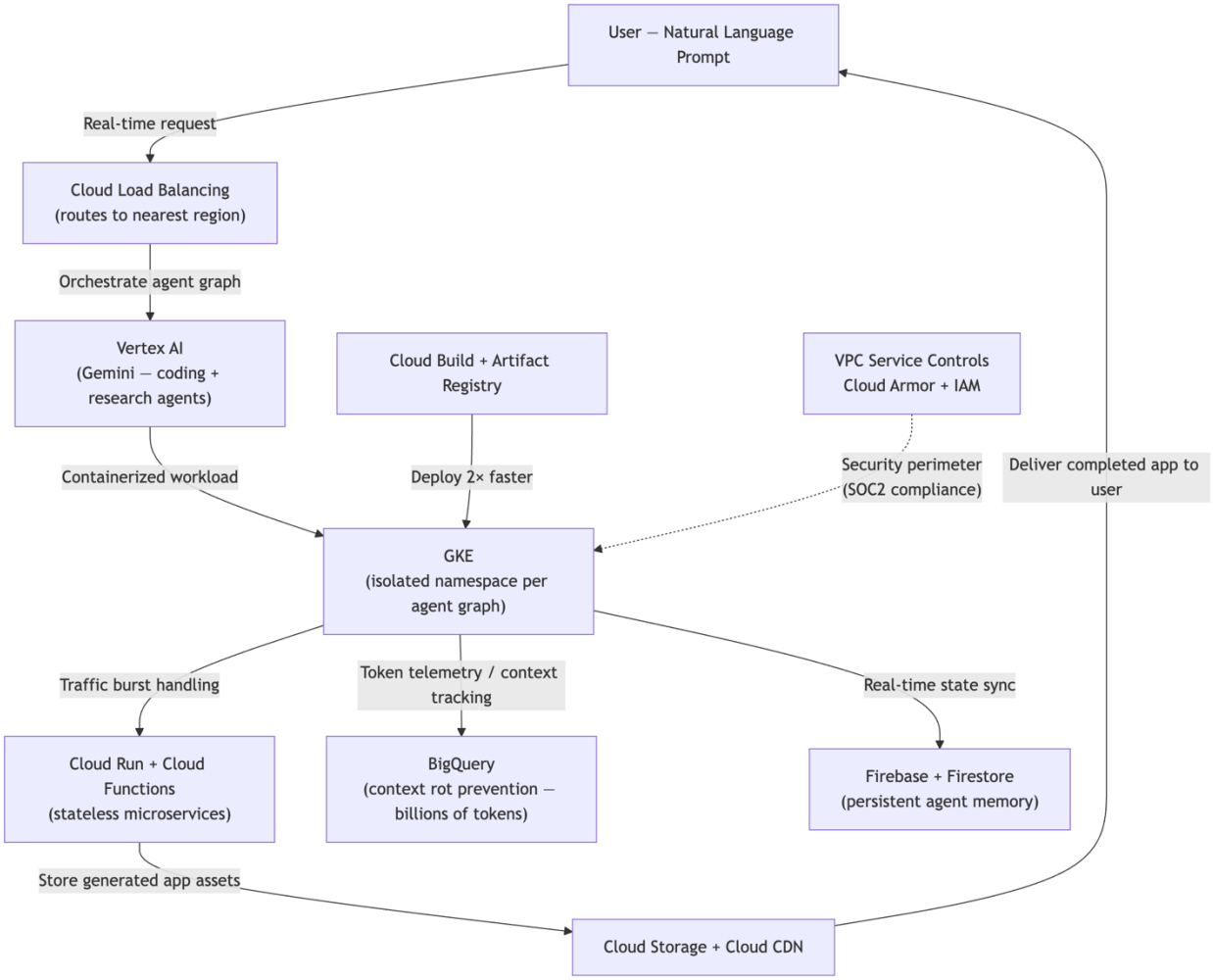

Figure 2: ChatAndBuild Google Cloud architecture — natural language prompt through Vertex AI orchestration, GKE execution, and CDN delivery to end user (Google Cloud, n.d.).

3.5. Reflection

ChatAndBuild's migration exemplifies NIST's **rapid elasticity** and **on-demand self-service** characteristics (Mell & Grance, 2011) at their most extreme. Scaling a worker fleet from 10 to 100+ Compute Engine instances in minutes – autonomously, based on load – is structurally impossible with on-premises infrastructure, where capacity must be physically provisioned days in advance (Google Cloud, n.d.). In the prior architecture, engineers acted as manual autoscalers; any instability meant broken agent sessions and eroded user trust. The 45% infrastructure overhead reduction reflects **measured service** in practice: consumption-based billing replaced fixed infrastructure costs, freeing the team to ship product rather than manage servers (Google Cloud, n.d.). Key ongoing risks include data sovereignty for AI-generated content across multi-region deployments and long-term vendor dependency on GCP's AI ecosystem.

4. Case Study 3 – Voith & AWS

4.1. Case Overview

Voith is a 150-year-old family-owned manufacturing company with over 22,000 employees across 60 countries, producing machinery for the hydroelectric, papermaking, and transportation industries (Amazon Web Services, n.d.). Its legacy IT footprint was vast and fragmented: 3,500 servers across 146 global locations managing applications that were difficult and expensive to maintain. Seeking to lower costs, improve productivity, and access modern cloud capabilities, Voith conducted a competitive provider assessment – evaluating

providers against criteria including service portfolio depth, global regional availability, and partner support capability (Kuijpers, 2022) – before selecting AWS based on its solution maturity and collaborative approach (Amazon Web Services, n.d.). The **migration proceeds in four structured waves, rehosting** (migrating workloads as-is), **consolidation**, **modernization**, **and retirement of legacy applications** (Linthicum, 2023) – supported by AWS Professional Services and Slalom as the AWS Implementation Partner.

4.2. Service Model Analysis

Voith's migration spans **IaaS** and **PaaS** service models, reflecting the phased, multi-wave approach to cloud adoption. In the initial rehosting wave, legacy applications are lifted and shifted to AWS virtual machine infrastructure (IaaS), maintaining the existing operating model while immediately achieving data center consolidation. As the migration progresses into modernization waves, Voith transitions workloads to managed PaaS services – such as Amazon Athena for serverless analytics – reducing the operational effort of running systems and eliminating the need for dedicated database administration staff (Amazon Web Services, n.d.).

Service Model	Characteristic	Fit for Victoria Police
IaaS	Virtual machines, storage; customer manages OS and runtime	Rehosting wave – lift-and-shift of 3,500 servers to AWS virtual infra
PaaS	Managed services; provider handles platform operations	Modernization wave – Amazon Athena, managed databases, DevOps tooling
SaaS	Fully managed software	Partial – AWS GameDays (training); future SaaS productivity tooling

Table 7: Service model analysis for Voith (Amazon Web Services, n.d.).

4.3. Deployment Model Comparison

Voith migrated to a **public cloud** model deployed across **six AWS regions**, replacing 146 globally dispersed on-premises locations. The choice of public cloud was driven by the need for global standardization, centralized security management, and access to modern managed services – objectives that a private or hybrid model could not deliver at the required global scale (Amazon Web Services, n.d.; McHaney, 2021). Infrastructure as Code (IaC) enforces consistent configuration standards across all regions, eliminating the policy fragmentation inherent in managing 146 independent sites.

Deployment Model	Cost	Standardization	Global Reach	Fit for Voith
On-Premises (146 sites)	High CAPEX – 3,500 servers	Low – fragmented legacy systems per site	Global but ungovernable	Prior state – high cost, poor standardization, legacy debt
Private Cloud	High CAPEX + Ops	Moderate	Limited to owned data centers	Cannot consolidate 60-country footprint at acceptable cost
Public Cloud (6 AWS regions)	OPEX – 30% projected savings	High – IaC enforces standards globally	6 regions cover global operations	**Adopted** – consolidation + modernization + cost reduction
Hybrid Cloud	Medium – dual infrastructure	Medium	High	Transitional state – may be required long-term for OT/SCADA system

Table 8: Deployment model comparison for Voith (Amazon Web Services, n.d.; McHaney, 2021).

4.4. Cloud Services Analysis

Voith's AWS engagement combines migration services, managed analytics, DevOps tooling, skills development infrastructure, and FinOps practices to deliver both immediate consolidation benefits and long-term operational modernization (Amazon Web Services, n.d.):

Service	Purpose	Business Outcome
AWS Professional Services	Expert-led migration planning and execution across four waves	Structured migration with increasing velocity as team skills develop wave-by-wavex
Slalom (AWS Implementation Partner)	On-the-ground migration execution and workforce upskilling	Accelerated migration with embedded partner expertise; reduced time-to-cloud
Amazon Athena	Serverless analytics – queries data in S3 without provisioning servers	Eliminates database server administration overhead; pay-per-query cost model
Infrastructure as Code (IaC)	Automated, version-controlled infrastructure provisioning through reusable scripts rather than manual configuration (Chapple, 2022)	Central quality standards maintained globally; eliminates manual configuration drift across 146→6 consolidation
Centralized DevOps Model	Operations previously scattered across business units moved to shared DevOps practice	Standardized deployment pipelines; reduced manual ops toil organization-wide
Centralized Security Management	Unified security governance across all AWS regions	Single security posture replaces 146 fragmented site policies; standardized compliance

AWS GameDays	Gamified learning events – hands-on AWS skills development for employees	Builds in-house cloud capability across 22,000+ employees; reduces external partner dependency
FinOps Practices	Cost governance – identify, optimize, and attribute cloud spend continuously	Tracks and validates 30% savings target; informs rightsizing and reserved capacity decisions

Table 9: AWS services utilized by Voith and their business outcomes (Amazon Web Services, n.d.).

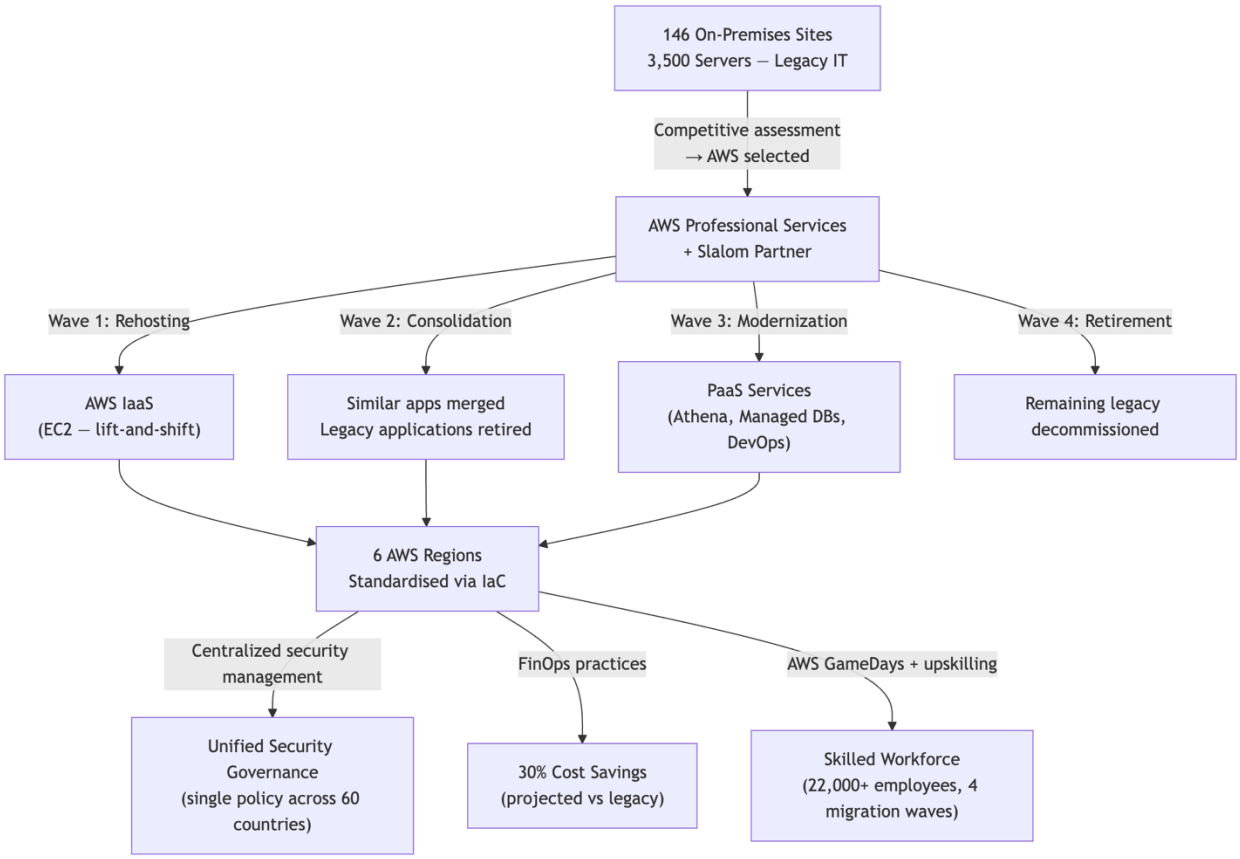

Figure 3: Voith AWS migration architecture — four-wave migration from 146 on-premises sites to 6 standardized AWS regions (Amazon Web Services, n.d.).

4.5. Reflection

Voith's case illustrates NIST's **resource pooling** and **measured service** characteristics at enterprise scale (Mell & Grance, 2011). Consolidating 3,500 servers across 146 locations into six AWS regions replaces fragmented hardware with shared, logically isolated cloud infrastructure – resource pooling at an organizational level. The projected 30% cost savings reflect measured service: Voith pays for consumption, with FinOps practices providing visibility to optimize spend continuously (Amazon Web Services, n.d.). The contrast with traditional IT is structural: each of 146 sites previously maintained its own servers and security policies, producing inconsistent governance across 60 countries. Cloud consolidation resolves this with a single IaC configuration enforcing standards globally and centralized security replacing fragmented site policies (McHaney, 2021). A residual risk is operational technology integration – manufacturing SCADA systems may require on-premises infrastructure long-term, creating a hybrid architecture requirement outside the current migration scope.

5. Conclusion

Across all three case studies, a consistent set of cloud principles underpinned the outcomes: on-demand self-service enabled each organization to provision capability without manual IT intervention; rapid elasticity allowed both ChatAndBuild and Voith to scale resources to match demand – whether a global hackathon spike or a 146-site migration wave; and measured service translated consumption directly into financial and operational accountability, from Victoria Police's governed fusion environments to Voith's 30% projected cost savings (Mell & Grance,

2011). Despite operating in radically different sectors – government law enforcement, AI startup, and industrial manufacturing – each organization achieved outcomes that were structurally impossible under their prior on-premises configurations, confirming that the essential cloud elements of automation are sector-agnostic even as their application is highly context-specific (McHaney, 2021).

In each case, fixed on-premises infrastructure represented a ceiling – on developer throughput (Victoria Police's constrained IT queue driving shadow application risk), on real-time AI generation capacity (ChatAndBuild's manual provisioning bottleneck degrading agent reliability), and on global standardization (Voith's fragmented 146-site estate producing inconsistent governance). Cloud removed these ceilings by decoupling capacity from physical hardware procurement and enabling OPEX-aligned, consumption-driven cost models in place of CAPEX-constrained, fixed-capacity traditional IT (McHaney, 2021; Mell & Grance, 2011).

Key areas warranting further investigation across these cases include cloud cost governance at enterprise scale, compliance frameworks for regulated sectors (law enforcement data residency, manufacturing OT system integration), and multi-cloud portability strategies as vendor dependency on Microsoft, Google Cloud, and AWS ecosystems deepens. The three case studies collectively demonstrate that cloud adoption is not a single transition event but an evolving capability requiring continuous governance, workforce skills development, and architectural adaptation to remain strategically effective.

6. Appendices

6.1. Appendix A – Cross-Case Comparison Summary

Attribute	Case Study 1 – Victoria Police	Case Study 2 - ChatAndBuild	Case Study 3 - Voith
Sector	Government / Law Enforcement	AI / Software Startup	Industrial Manufacturing
Cloud Provider	Microsoft (Azure ecosystem)	Google Cloud Platform (GCP)	AWS (Amazon Web Services)
Service Model	PaaS (primary) + SaaS (partial)	IaaS + PaaS	IaaS + PaaS
Deployment Model	Hybrid – multi-environment (personal → fusion → enterprise)	Public cloud – multi-region	Public cloud – 6 AWS regions
Prior State	Ungoverned MS Access shadow apps; central IT bottleneck	30-70% higher latency; engineers manually provisioning serves days before events	3,500 servers across 146 locations; fragmented legacy applications
Key Services	Power Apps, Dataverse, Power Platform Environment, Power Automate, Copilot Studio	Vertex AI, GKE, Cloud Run, BigQuery, Firebase, Cloud CDN, Cloud Build, Cloud Armor	AWS Professional Services, Amazon Athena, IaC, Centralized DevOps, AWS GameDays, FinOps
Key Primary Automation Outcome	Governed fusion framework – citizen developers build enterprise-grade apps via tiered approval	Real-time AI code and agent generation at sub-second latency for 140,000+ users	Consolidated global operations; IaC-enforced standards across 60 countries
Headline Result	2 enterprise apps in production (Global Whereabouts; Bail Reporting)	45% infrastructure overhead ↓; 30-70% latency ↓; 2x faster feature delivery	30% projected cost savings; 146 to 6 site consolidation

Table A1 –Side-by-side comparison of the three case studies across provider, service model, deployment model, key services, and business outcomes.

6.2. Appendix B – NIST Essential Characteristics, Cross-Case Mapping

NIST Characteristic	Case Study 1 – Victoria Police	Case Study 2 - ChatAndBuild	Case Study 3 - Voith
On-demand self-service	Primary - police personnel provision their own Power Platform environments and build applications without central IT intervention (Microsoft, 2026)	Autoscaling policies provision and terminate Compute Engine instances automatically without engineer intervention (Google Cloud, n.d.)	IaC enables automated, consistent provisioning of infrastructure across 6 regions without manual configuration per site (Amazon Web Services, n.d.)
Broad network access	Officers across all stations access Global Whereabouts and Bail Reporting from any device via Microsoft 365 (Microsoft, 2026)	140,000+ users globally access the platform via browser; Cloud Load Balancing routes to the nearest region for sub-second response (Google Cloud, n.d.)	22,000+ employees across 60 countries access centralized systems via 6 standardized AWS regions (Amazon Web Services, n.d.)
Resource pooling	Multiple business units share Power Platform infrastructure; Dataverse pools structured data storage across all fusion applications (Microsoft, 2026)	GKE clusters pool compute across thousands of concurrent agent graphs and user sessions (Google Cloud, n.d.)	Primary driver - 3,500 servers across 146 sites pooled into 6 shared, logically isolated AWS regions (Amazon Web Services, n.d.)
Rapid elasticity	Limited - Power Platform environments scale within Microsoft's managed bounds; not the primary driver for this case	Primary driver - worker fleet scales from 10 → 100+ Compute Engine instances within minutes during global	Wave-based migration progressively expands the cloud footprint; real-time elasticity is not the primary driver for this manufacturing workload

		hackathon events with zero downtime (Google Cloud, n.d.)	
Measured service	Tiered governance tracks app usage and adoption; Power Platform visibility surfaces which app reach 300+ users and warrant promotion (Microsoft, 2026)	45% infrastructure overhead reduction reflects consumption-based billing replacing fixed legacy infrastructure costs (Google Cloud, n.d.)	Primary driver – FinOps practices provide continuous cost visibility and track progress toward 30% savings target; OPEX model replaces CAPEX-heavy 146-site estate (Amazon Web Services, n.d.)

Table B1: Mapping of NIST's five essential cloud characteristics (Mell & Grance, 2011) to each case study, showing which characteristics were primary drivers of the outcomes achieved.

End of Appendix Section

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